

SELF-AUDITING TRADING AGENT · ALPACA

Vermilion

An autonomous AI that wakes itself up, scores the market, and refuses by default.

Prasanna · lablab.ai × Alpaca AI Trading Agents Hackathon · Aug 2026

Paper · Alpaca

DeepSeek

MCP · 17 tools

THE PITCH

Refuse by default. Ask by exception.

Vermilion is an autonomous trading agent on Alpaca. It wakes itself up, evaluates your watchlist against the live market, and writes down every decision it takes — traded or refused. If confidence is too low, Vermilion says no, and tells you why.

AUTONOMOUS

Self-running cycle, no human needed to start it.

AUDITABLE

Every decision persisted with reasoning + sources.

HUMAN-IN-THE-LOOP

Every order waits for your explicit approval.

THE AGENT LOOP

One cycle. Every 15 minutes. Or on demand.

01

WATCH

Pull live snapshots for the 13-symbol watchlist.

02

ASK

DeepSeek scores every name, refuses by default.

03

PERSIST

Write the decision + reasoning to the audit log.

04

QUEUE

Push the order to Telegram, WhatsApp, or Email for human approval.

3 triggers: cron · wake-on-visit · manual button

RISK GATES

Eight gates. The agent can refuse, but cannot escape.

60%

Confidence floor

Below 60%: refuse.

8%

Per-position cap

No single name > 8% of equity.

1k\$

Order notional

Default cap on market orders.

10_m

Live-quote sanity

Refuse stale snapshots.

RTH

Market-closed check

No day orders outside hours.

20%

Drawdown breaker

Pause if equity drops 20%.

HITL

Human approval

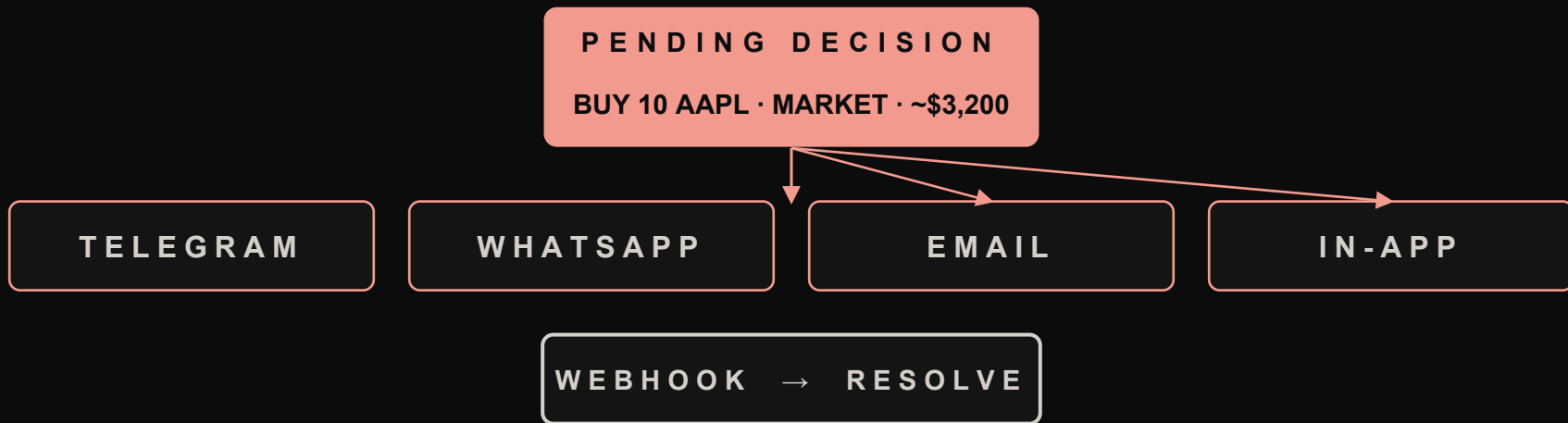
Always. No auto-execute.

RLS

Row-level security

No cross-tenant reads.

Four channels. One truth. First response wins.



Approve · decline · comment with free text. First response closes the row.

OPTIONS

Four strategies. Composed live from the option chain.

COVERED CALL

Sell 1 call per 100 shares. Caps upside, collects premium.

PROTECTIVE PUT

Buy 1 put per 100 shares. Floors the loss, costs premium.

BULL CALL SPREAD

Long lower call + short higher call. Defined risk.

BEAR PUT SPREAD

Long higher put + short lower put. Profits on drop.

OCC-symbol legs, live greeks, IV. Pushes into the same HITL queue.

17 tools. Any MCP host can drive the broker.

alpaca_get_account	alpaca_get_positions	alpaca_get_open_orders
alpaca_cancel_order	alpaca_get_snapshot	alpaca_get_quotes
alpaca_get_news	alpaca_get_bars	alpaca_get_portfolio_history
alpaca_get_clock	alpaca_get_asset	alpaca_place_order
alpaca_get_option_expirations	alpaca_get_option_chain	alpaca_get_option_snapshots
alpaca_place_option_order	alpaca_compose_strategy	

ARCHITECTURE

The same broker on the web, on cron, and in any MCP host.

FRONTEND

Next.js 16 · React 19 · Tailwind v4 · Perplexity-style chat · Obsidian UI



AGENT CORE

Autonomy · constitution · queue · lifecycle · DeepSeek tool calls



BROKER

Alpaca Trading + Market Data · MCP server · 14 Supabase tables · RLS

Single source of truth: lib/alpaca/server.ts. Every other path goes through it.

BY THE NUMBERS

What shipped.

13

symbols in the watchlist

8

hard risk gates

4

notification channels

17

MCP tools, zero npm
deps

STACK

Next.js 16 · React 19 · TypeScript · Tailwind v4 · Supabase · DeepSeek · Alpaca Trading API · Alpaca Market Data · MCP

T H A N K S

Vermilion shows up. Looks at the market. Says no.

Refuse by default. Ask by exception. Audit everything.

Live demo vermilion-mu.vercel.app

Source github.com/Prasannaverse13/Vermilion-

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